

Math Formulation: Constraining a Hacked Reward — Gates, Aggregation, and Why In-Loop Alerts Fail

S Hariharan, J Shrenik Reddy, Ansa N

This continues from `open_items.pdf`, which formalized the three items left open by the working draft. Section numbers continue from 10; equation numbers continue from (16). Where that document specified the *algorithm*, this one specifies the *constraints* added after the algorithm was found to be exploitable, and states precisely what each does and does not guarantee. Every claim marked “measured” is reproduced in `documentation/reward_hacking_investigation.md`.

1 The validity predicate as the place to spend a hard constraint

Section 4 defined $\text{Valid}(m)$ as RDKit sanitization success, and Eq. 5 collapses invalid molecules to a flat reward r_{inv} . Write the sanitization predicate as $\text{Valid}_0(m)$ and generalize:

$$\text{Valid}(m) = \text{Valid}_0(m) \wedge \mathbb{K}[\text{HAC}(m) \geq h_{\min}], \quad (1)$$

with HAC the heavy-atom count. Everything downstream is unchanged: gated molecules take r_{inv} and the hurdle advantage floor of Section 9, exactly as a valence violation does.

Why a gate and not a term. Two arguments, and they are different. First, *monotonicity*: any reward term increasing in molecular size has its optimum at unbounded size, so a soft size reward is hackable in the opposite direction. A hinge $\max(0, h_{\min} - \text{HAC})$ avoids that, but—second—*gradient conditioning*: a term intended to drive the reward to zero must approach zero, and under the multiplicative aggregation of Section 2 the sensitivity $\partial G / \partial S_k \propto G / S_k$ diverges there. Eq. (1) has no gradient to climb and none to explode. The constraint is spent where it is free.

The cost, and the condition under which it is not free. Gating removes candidates from the group. Writing n_k for the number of valid molecules at step k , Section 9 established that the group-relative advantage carries no information when $n_k < 2$, and returns exactly zero. So Eq. (1) is safe only while

$$\Pr [n_k \geq 2] \approx 1, \quad (2)$$

which must be monitored rather than assumed. Measured at $h_{\min} = 10$ and $N = 64$: $n_k \in [61, 62]$ on average, and the gated fraction falls from $\approx 9\%$ at initialization to 2–4% by step 150. That decline is the diagnostic signature of a hard gate: the policy learns to avoid it rather than paying it repeatedly, which a soft penalty would not produce.

2 Multiplicative aggregation

Eq. 5 aggregates the reward terms linearly. A weighted sum permits compensation: a term at its maximum can outbid another near its minimum, which is precisely how a molecule scoring

$C = 0.999$ and QED = 0.47 outranks one scoring $C = 0.811$ and QED = 0.54. Replace the sum with a weighted geometric mean over terms $S_k \in [0, 1]$:

$$G(m) = \left(\prod_k S_k(m)^{w_k} \right)^{1/\sum_k w_k} = \exp\left(\frac{1}{\sum_k w_k} \sum_k w_k \log S_k(m)\right). \quad (3)$$

The right-hand form is what should be implemented: it is stable, and it makes the failure mode explicit, since $\log S_k \rightarrow -\infty$ as $S_k \rightarrow 0$. Clipping S_k to $[\varepsilon, 1]$ bounds both the value and the sensitivity.

Two properties follow, and only the first is usually stated.

1. **Veto.** $G \rightarrow 0$ if any $S_k \rightarrow 0$, regardless of the others. This is the intended behaviour.
2. **Dilution.** The exponent $w_k/\sum_j w_j$ attenuates every factor. A term contributing S_k changes G by $S_k^{w_k/\sum w}$, so with five equally weighted terms a factor of 0.55 becomes 0.89. A penalty calibrated on S_k in isolation will therefore be far weaker in aggregate than intended, and must be calibrated on its exponentiated contribution instead.

Property 2 is why Section 1 is a gate rather than a factor: a veto expressed multiplicatively is diluted, a veto expressed through Valid is not.

Scale compatibility. Eq. (3) requires $S_k \in [0, 1]$. Any monotone recalibration of a term that leaves that interval—such as a logit transform centred on a reference mean, which is negative below it—is incompatible, and the incompatibility is structural rather than data-dependent: whether a negative value appears in a given group is a property of the sample, not of the configuration.

3 Why an in-loop substructure penalty does not generalize

Let $\mathcal{A}$ be a finite set of substructure patterns and

$$N_{\mathcal{A}}(m) = \sum_{a \in \mathcal{A}} \mathbb{1}[a \subseteq m], \quad S_{\mathcal{A}}(m) = e^{-\lambda N_{\mathcal{A}}(m)}, \quad (4)$$

with $a \subseteq m$ denoting a substructure match. Adding $S_{\mathcal{A}}$ to Eq. (3) penalizes members of $\mathcal{A}$ and *nothing else*.

This is the crux. The quantity of interest is chemical reactivity, call it $R^*(m)$, which is unobserved. $N_{\mathcal{A}}$ is a proxy, and the policy optimizes the proxy. Because $\mathcal{A}$ is finite and the space of reactive motifs is not, there exist molecules with $N_{\mathcal{A}}(m) = 0$ and $R^*(m)$ large, and the optimizer has a gradient toward exactly those. Formally, optimizing $S_{\mathcal{A}}$ reduces $\mathbb{E}[N_{\mathcal{A}}]$ under π_{θ} with no implication for $\mathbb{E}[R^*]$ — the two are related only through whatever correlation $\mathcal{A}$ happens to carry, and the optimization is free to destroy it.

The held-out instrument criterion. Since R^* is unobservable, the only available test is a second pattern set $\mathcal{B}$ with $\mathcal{B} \cap \mathcal{A} = \emptyset$, never used in training. An intervention is credited only if

$$\mathbb{E}_{\pi_{\theta}}[N_{\mathcal{B}}] < \mathbb{E}_{\pi_{\theta_{\text{base}}}}[N_{\mathcal{B}}] \quad (5)$$

by a margin exceeding the seed-to-seed variation of $N_{\mathcal{B}}$. Measured at $\lambda = 0.3$, $w_{\mathcal{A}} = 3$: the trained-against rate fell $0.339 \rightarrow 0.203$ while BRENK moved $0.438 \rightarrow 0.406$ and BRENK+PAINS+NIH $0.464 \rightarrow 0.427$, both within seed noise. Eq. (5) is not satisfied.

Corollary. Training against $\mathcal{A} \cup \mathcal{B}$ to close the gap is self-defeating: it leaves no set disjoint from the training signal, so Eq. (5) becomes uncomputable and the failure becomes undetectable rather than absent. At least one instrument must be permanently withheld.

4 Post-hoc selection as a distinct operator

The argument of Section 3 turns on $\mathcal{A}$ entering the objective. It therefore does not apply to a filter applied after training. Let π_θ be the trained policy and define the selection operator

$$\mathcal{F}[\pi_\theta](m) = \frac{\pi_\theta(m) \mathbb{1}[N_{\mathcal{A} \cup \mathcal{B}}(m) = 0]}{\Pr_{\pi_\theta}[N_{\mathcal{A} \cup \mathcal{B}} = 0]}, \quad (6)$$

the policy conditioned on passing the union of every available instrument. Since θ is fixed when $\mathcal{F}$ is applied, no gradient and no reward reaches $\mathcal{A} \cup \mathcal{B}$, and the union may be used in full — the constraint of Section 3 is lifted exactly because nothing is being trained.

What Eq. (6) buys and what it does not:

1. It is not gameable, by construction.
2. It does not change π_θ . The generator remains as reactive as before; the denominator is the yield, measured at 0.40.
3. It is not a safety guarantee. $\Pr[N = 0]$ is bounded below 1 even for approved drugs — measured, 39% of the BBB+ molecules in BBBP fail the same union — so the predicate is neither necessary nor sufficient for safety, only for the absence of *recognized* motifs.

5 A stopping criterion from the reward’s own uncertainty

Section 10.3 tracked $\text{Var}_i[D_{\phi_k}(m_i)]$ as a diagnostic for the discriminator. The frozen classifier admits a sharper statement, because its uncertainty can be estimated directly.

Let $\ell(m) = \text{logit } C(m)$ and let $\tilde{C}^{(1)}, \dots, \tilde{C}^{(K)}$ be K stochastic forward passes with dropout active and normalization layers held in inference mode — the latter being required, or C ceases to be a function of m alone and the stationarity argument of Section 3.1 fails. Define

$$\sigma_{\text{sig}}(k)^2 = \text{Var}_{i \in \mathcal{G}_k}[\bar{\ell}(m_i)], \quad \sigma_{\text{noise}}(k)^2 = \mathbb{E}_{i \in \mathcal{G}_k}[\text{Var}_j \tilde{\ell}^{(j)}(m_i)], \quad (7)$$

the between-molecule spread of the consensus logit and the mean within-molecule spread across dropout samples, over group $\mathcal{G}_k$. The first is what the group-relative advantage ranks on; the second is the model’s own epistemic uncertainty about each ranking. Their ratio

$$\text{SNR}(k) = \sigma_{\text{sig}}(k) / \sigma_{\text{noise}}(k) \quad (8)$$

is therefore the quantity that decides whether the reward is still informative. When $\text{SNR}(k) < 1$ the differences the advantage ranks on are smaller than the classifier’s uncertainty about them, and further optimization against C pursues noise.

Measured on the baseline: SNR falls $2.52 \rightarrow 1.24 \rightarrow 0.42$ at steps 0, 100, 199, crossing 1 between steps 100 and 150 — the same interval over which scaffold diversity collapses.

Two consequences. First, σ_{noise} is approximately constant in k (measured: 0.90–0.98 throughout) while σ_{sig} falls by $5.5\times$, so Eq. (8) is driven almost entirely by signal loss rather than by the model becoming less certain. Second, the apparent saturation of C in probability space is an artifact of the sigmoid: the same MC-dropout spread is flat on the logit scale (0.86 on real molecules,

0.99 on collapsed samples) while appearing to shrink $3.6\times$ in probability. A recalibration therefore restores σ_{sig} but leaves σ_{noise} untouched — it makes the reward legible, not more informative, and Eq. (8) is the statement of that distinction.

6 What none of this addresses

Each constraint above narrows the policy’s access to a particular exploit. None changes the objective. Writing $y(m)$ for the BBBP label, the classifier estimates

$$C(m) \approx \Pr[y(m) = 1] = \Pr[m \text{ crosses the blood-brain barrier}], \quad (9)$$

which is a statement about transport and carries no term for therapeutic intent. The training set is consistent with that reading: of the 65 molecules with $\text{HAC} \leq 8$ in BBBP, 62 are labelled positive, and they include chloroform, dichloromethane, bromoform and nitrous oxide. Carbon tetrachloride is therefore not an extrapolation failure of C but an interpolation between positive training examples, and C is correct about the only question it was asked.

Any constraint added to Eq. 5 is a counterweight to that objective rather than a correction of it. Five have now been tested — property-distribution distance, applicability domain, Tox21 activity, in-loop substructure alerts, and multiplicative aggregation — and only the size gate and post-hoc selection survive their own validation. The remaining intervention that acts on the objective itself is curation of the label set, and it has not been attempted.